EXPERIMENT 5: GENERATE TEXT USING A PRE-TRAINED LARGE LANGUAGE MODEL

Aim

To generate text using a pre-trained Large Language Model and analyze the generated response for relevance, coherence, and fluency.

Algorithm

1. Import the text-generation pipeline from Transformers.
2. Load a pre-trained GPT-2 language model.
3. Provide an input prompt.
4. Generate new text using the model.
5. Display the generated response.
6. Analyze the response for relevance, coherence, and fluency.

Program

```
# Experiment 5: Text Generation Using GPT-2

from transformers import pipeline

# Load the pre-trained GPT-2 model
generator = pipeline(
    "text-generation",
    model="openai-community/gpt2"
)

# Input prompt
prompt = "Artificial intelligence is useful because"

# Generate text
result = generator(
    prompt,
    max_new_tokens=50,
    num_return_sequences=1,
    do_sample=False,
    pad_token_id=generator.tokenizer.eos_token_id
```

```
)

# Extract generated text
generated_text = result[0]["generated_text"]

# Display output
print("Input Prompt:")
print(prompt)

print("\nGenerated Text:")
print(generated_text)

print("\nResponse Analysis:")
print("Relevance: Check whether the text relates to the prompt.")
print("Coherence: Check whether the sentences make sense.")
print("Fluency: Check whether the text reads naturally.")
print("Factual Accuracy: Verify factual claims independently.")
```

Input

Artificial intelligence is useful because

Expected Output

The following is an illustration, not a guaranteed model response.

```
Input Prompt:
Artificial intelligence is useful because

Generated Text:
Artificial intelligence is useful because it can help
people solve problems, process information and automate
repetitive tasks in different fields.

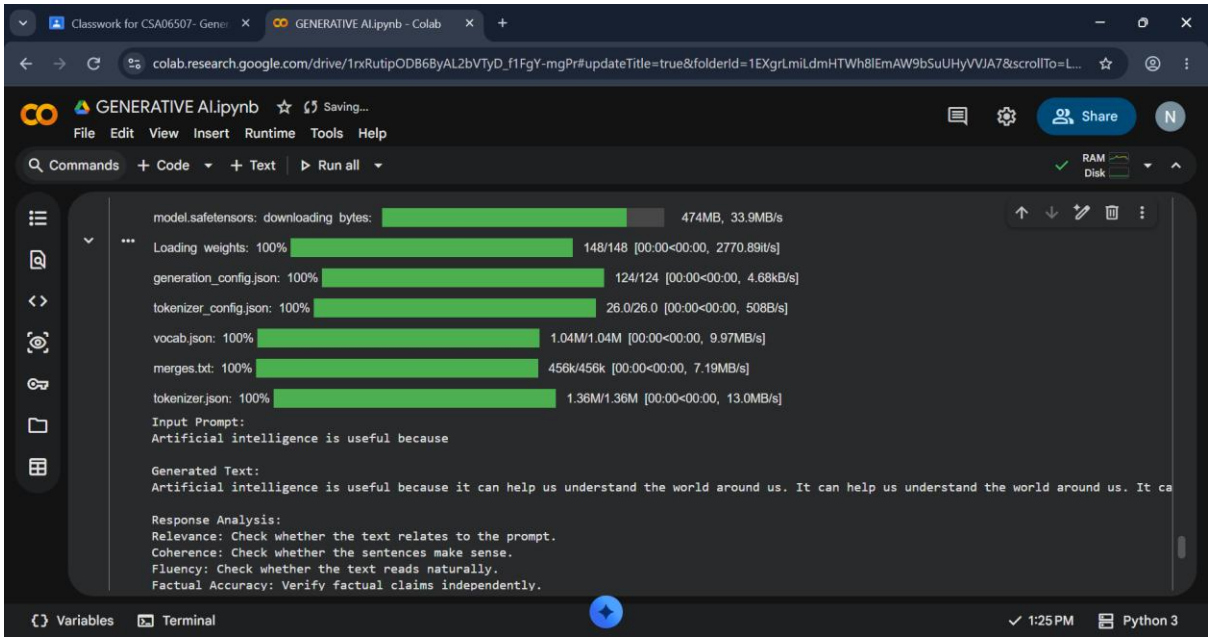
Response Analysis:
Relevance: Check whether the text relates to the prompt.
Coherence: Check whether the sentences make sense.
Fluency: Check whether the text reads naturally.
Factual Accuracy: Verify factual claims independently.
```

The actual GPT-2 response may differ and may not be as clear as the illustrative output.

Analysis of Generated Response

Criterion	Description
Relevance	Determines whether the generated text addresses the prompt
Coherence	Checks whether the text is logically connected
Fluency	Checks grammar and readability
Factual accuracy	Checks whether the claims are correct
Hallucination	Identifies unsupported or fabricated information

Note: GPT-2 is a relatively small, older language model. It is suitable for demonstrating text generation, but it may produce incorrect, repetitive, or incomplete responses.



Result

Thus, text was generated using a pre-trained GPT-2 language model, and the response was analyzed for relevance, coherence, fluency, and factual accuracy.